SRv6-based Rate Control

draft-lx-spring-srv6-rate-control-00
draft-xz-6man-rate-option-00
draft-xz-rtgwg-srv6-rate-notification-00

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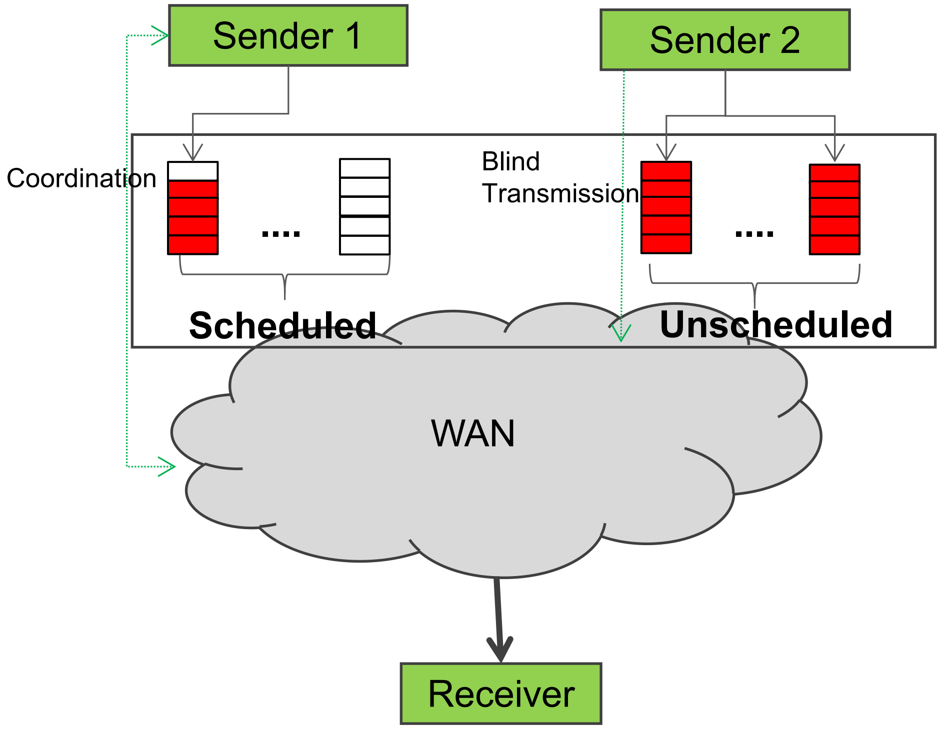
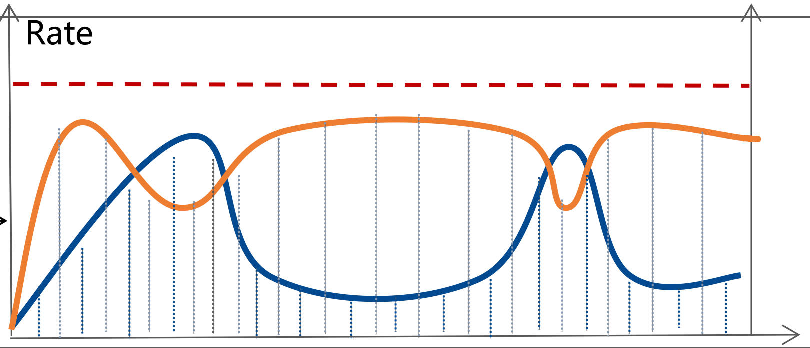
Problems for Network Congestion and High-latency

- Problem1 : The lack of coordination generates a large volume of unplanned elephant flow transmissions in the network, which causes severe WAN congestion and packet loss.



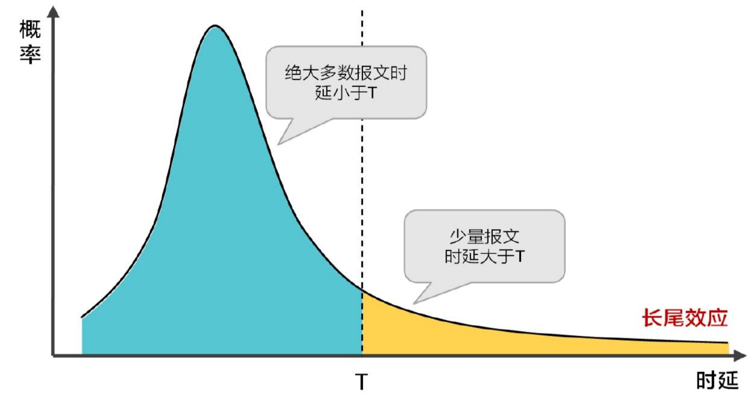
- Problem 2 : Traditional host-driven congestion control lacks network-aware rate control, leading to passive competition among multiple flows and FCT fluctuation.

AI Traffic

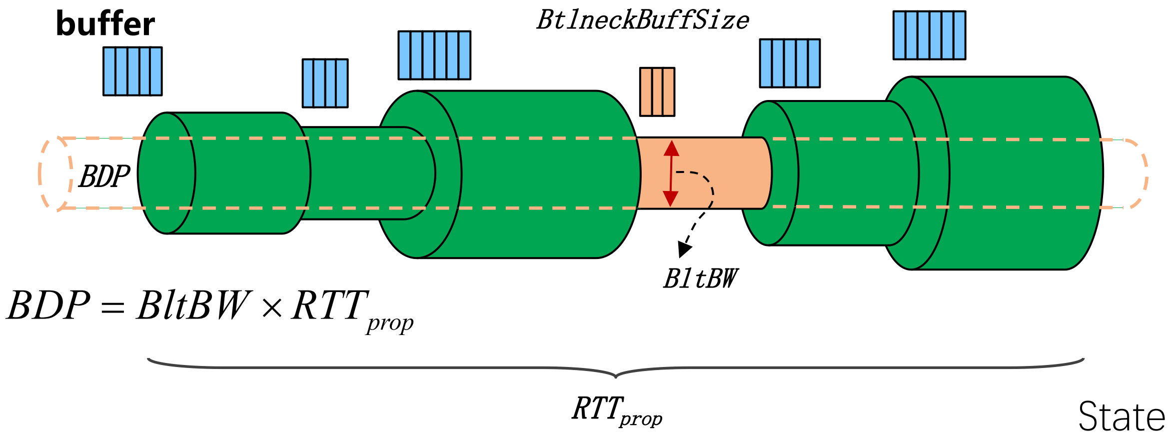


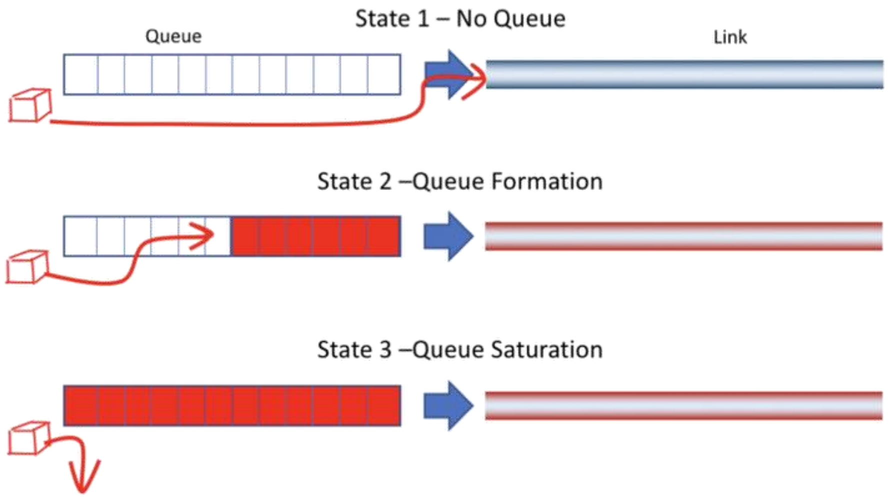
- Problem 3： Continuous accumulation of packets on the bottleneck link leads to the long-tail effect, triggering queuing delay and congestion-induced packet loss.

Long-tail effect



Network States for Evolving AI Workload





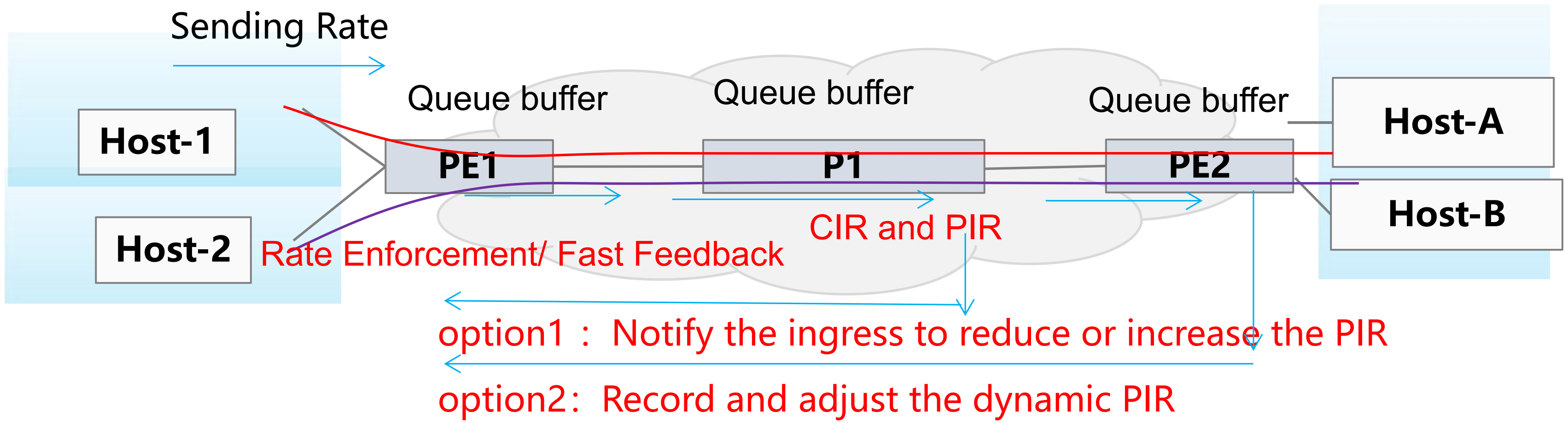
State 1： inflight∈(0, BDP]
State 2： inflight∈(BDP, BDP+BtlneckBuffSize]
State 3： inflight∈(BDP+BtlneckBuffSize, ∞)

BDP:(Bndwidth-Delay Product)，The maximum number of bits that a link can accommodate.
inflight: (in-flight data): Data that has been sent by the sender but has not yet received an acknowledgment from the receiver.

State 1: No Queuing
When there is little data on the link, all data is transmitted directly on the sending link with no queued packets. In this case, the latency is the lowest. But the network bandwidth is underutilized.
State 2: Queuing Occurs
As more data is transmitted, packets start to queue, and the latency begins to increase. This is the best case with optimal range {BDP,BDP+BtneckBuffSize} which can be transformed to sending rate range.
State 3: Queue Saturation
The system enters the third state when the queue is full, and packet loss occurs.

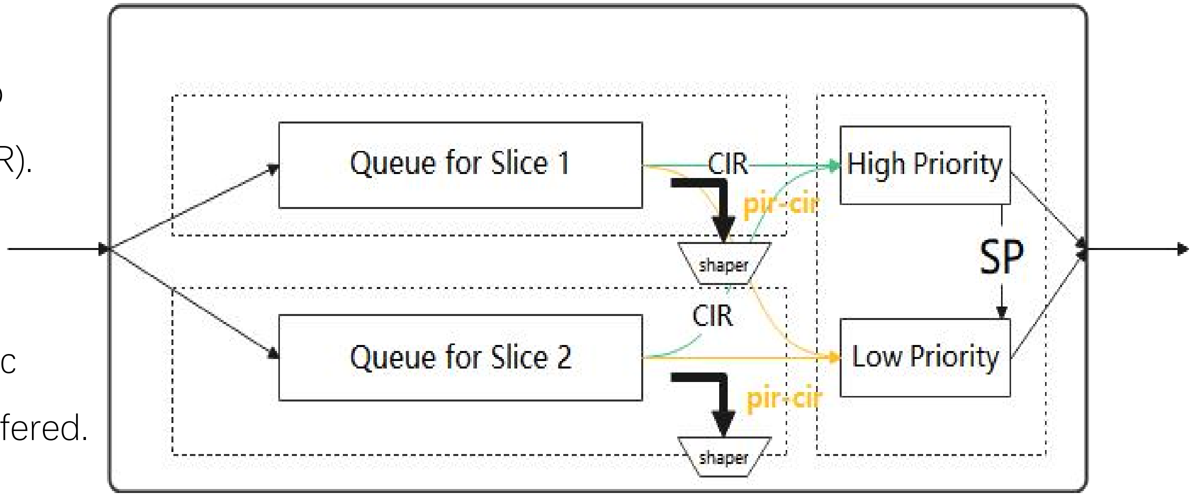
CIR and PIR in SRv6 Network

- Initial Configuration of CIR and PIR and Dynamic PIR Adjustment in SRv6 Network
  - When an SRv6 network slice is oversubscribed, the traffic allocated to a specific slice can exceed its reserved or committed resources (such as bandwidth, queues, etc.).
  - The configuration of CIR and PIR is the key to ensuring efficient network resource utilization and rigid SLA guarantee.
  - The configuration of CIR and PIR is closely related to the parameters (queue bandwidth, buffer) in SRv6 slices, and the characteristics of AI/HPC service traffic models (such as average rate, burst period, packet size). It is necessary to dynamically control CIR and PIR reasonably to avoid over-reservation of slice resources or congestion risks.
  - *Control the rate to keep the inflight data within the optimal range.*



SRv6-based Rate Control

- draft-lx-spring-srv6-rate-control-00
- Token-Based Queue Scheduling for SRv6-based Slices
  - The token bucket scheduler assigns high-priority tokens to traffic up to the CIR and low-priority tokens to traffic exceeding the CIR (up to the PIR).
  - Queue resources (bandwidth) can be shared if CIR tokens for a specific queue are not fully consumed.
  - When CIR traffic in a queue requires scheduling, high-priority CIR traffic
  - MUST be processed first and low-priority PIR traffic is temporarily buffered.



- Initial Rate Setting
  - The CIR is the committed, guaranteed rate per the service contract. The initial PIR can be calculated based on several factors to allow for efficient burst accommodation without causing congestion.
  - Initial PIR = CIR + (maxBufferSize/RTT+maxLinkBw - sumCIR) * p
- Dynamic Rate Adjusting
  - This dynamic adjustment enables coordinated resource allocation across the network, helping to prevent localized congestion from spreading.
    - Option 1: A transit monitors the queue status and generates a rate notification to trigger a PIR update (e.g., rate up or down between {CIR, Initial PIR}).
    - Option 2 :A ingress tracks the minimum PIR discovered along the forward path and use this value to update the effective bottleneck PIR.

Rate Notification using ICMPv6/UDP

- draft-xz-rtgwg-srv6-rate-notification-00
  - It enables a transit or egress node to dynamically <u>notify the ingress (headend) node about a recommended rate range (MinRate and MaxRate)</u> when localized congestion is detected or when underutilized bandwidth is identified,allowing the headend to perform proactive traffic shaping and rate enforcement or send fast feedback to the clients.
- Rate-Down Notification
  - when conditions indicate that the impending congestion are met for a specific slice or queue (e.g.,75% of queue depth or sustained high queueing delay)
- Rate-Up Notification
  - when conditions indicate that bandwidth is underutilized along the SRv6 path (e.g., queue buffer is not occupied and the link utilization is lower than 80%)

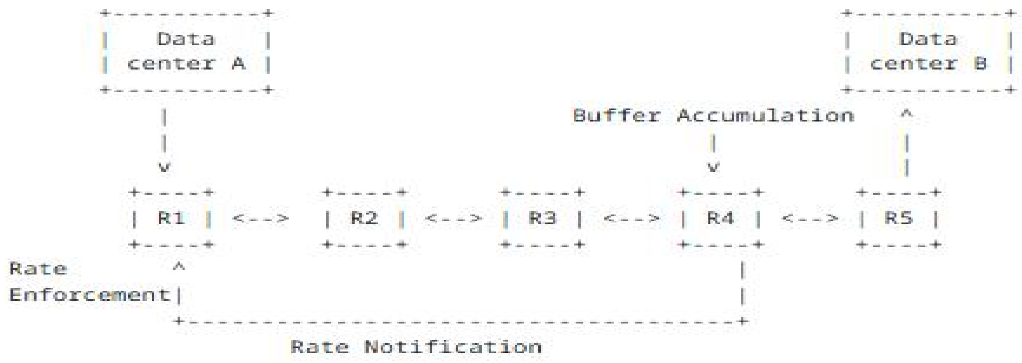


Figure 1: Rate Notification in SRv6 Network

```
 0                   1                   2                   3
 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|  Type=TBD1    |     Code      |          Checksum             |
+---------------+---------------+-------------------------------+
|R|   Flags     |   Priority    |          Reserved             |
+---------------+---------------+-------------------------------+
|                            MinRate                            |
+---------------------------------------------------------------+
|                            MaxRate                            |
+---------------------------------------------------------------+
|                            Slice ID                           |
+---------------------------------------------------------------+
```

```
 0                   1                   2                   3
 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|      UDP source port          |  UDP destination port=TBD2    |
+-------------------------------+-------------------------------+
|          UDP length           |         UDP checksum          |
+---------------+---------------+-------------------------------+
|     Flags     |   Priority    |          Reserved             |
+---------------+---------------+-------------------------------+
|                            MinRate                            |
+---------------------------------------------------------------+
|                            MaxRate                            |
+---------------------------------------------------------------+
|                            Slice ID                           |
+---------------------------------------------------------------+
```

Rate Discovery using IPv6 HBH/DOH

- draft-xz-6man-rate-option-00
  - It enables minimum rate limit discovery along the forward path and each node along the path can update the option with the minimum of its local Maximum Rate (MRate) and the recorded value.
  - The head node sends the IPv6/SRv6 packets, recording the minimum MRate of nodes along the path. e.g.,the PIR of each node is:
    - PIR(h)= CIR + (PIR(h-1) - CIR) * a;
  - Each node compares the PIR with the MRate in the packet, replaces it with the minimum value, and the tail node obtains the Path MRate:
    - Path MRate = min { PIR(h) }, for h = 1, 2, ... N;

```
+-----------+                                                 +------------+
| Source    |                                                 |Destination|
| Host      |                                                 |Host        |
+-----------+                                                 +------------+
      |                                                              ^
      |                                                              |
      v                                                              |
  +-----+       +-----+       +-----+       +-----+       +-----+
  | R1  | <---> | R2  | <---> | R3  | <---> | R4  | <---> | R5  |
  +-----+       +-----+       +-----+       +-----+       +-----+
    ---->20Gbps ----> 10Gbps---->5Gbps---->10Gbps---->
                                                             |
  <----------------------------------------------------------+
                      Minimum MRate (e.g.,5Gbps)
```

Figure 1: An Example IPv6/SRv6 Path between the Source Host and the Destination Host

```
+---------------+---------------+---------------+---------------+
|  Option Type  |Option Data Len|R|    Flags    |    Priority   |
+---------------+---------------+-------------------------------+
|                         Minimal MRate                         |
----------------------------------------------------------------+
|                            Slice ID                           |
+---------------------------------------------------------------+
```

Figure 2: Format of the Minimum MRate Hop-by-Hop Option

- **Thanks !**
- **Comments and suggestions are welcome.**